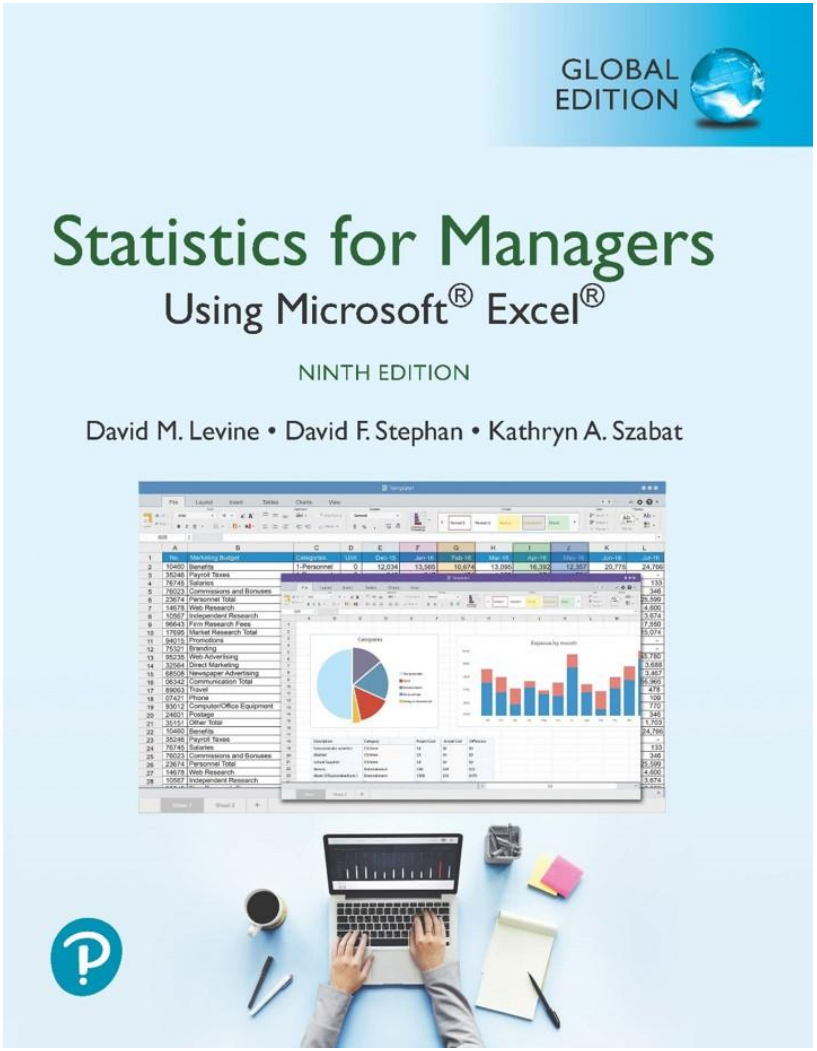


Chapter 13

Simple Linear Regression

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918 · 2018

Copyright © Pearson Education Ltd.

Objectives

In this chapter, you learn:

- To understand the meaning of the regression coefficients b_0 and b_1.
- To understand the usefulness of regression analysis and how to properly perform regression analysis.

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918 · 2018

Copyright © Pearson Education Ltd.

Regression Analysis Techniques Help Uncover Relationships Between Variables

DCOVA

- Regression analysis is used to:
  - Predict the value of a dependent variable based on the value of at least one independent variable.
  - Explain the impact of changes in an independent variable on the dependent variable.

Dependent variable: the variable we wish to predict or explain.

Independent variable: the variable used to predict or explain the dependent variable.

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918 · 2018

Copyright © Pearson Education Ltd.

Preliminary Analysis – Scatter Plots

DCOVA

- A scatter plot can be used to:
  - Visualise the relationship between X and Y variables.
  - Help suggest a starting point for regression analysis.

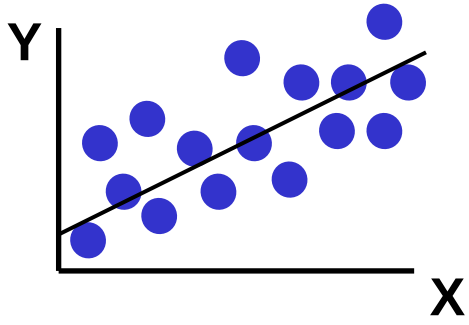
UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918 · 2018

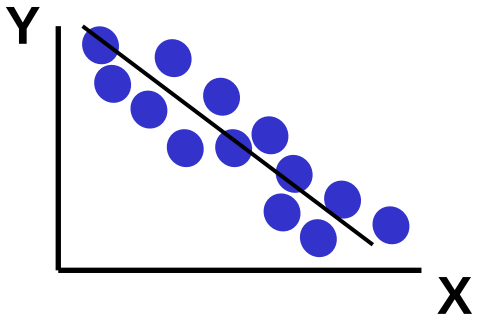
Copyright © Pearson Education Ltd.

Types of Relationships

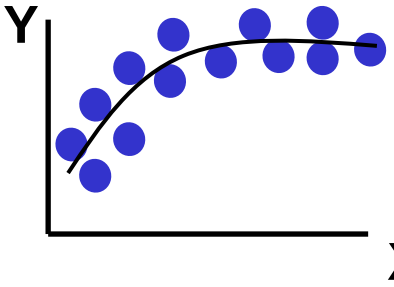
DCOVA

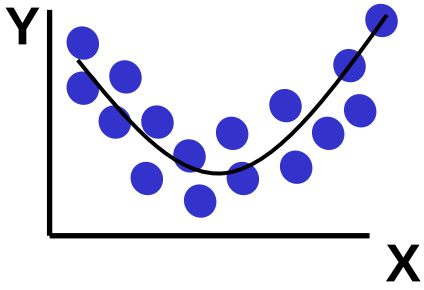
Linear Relationships

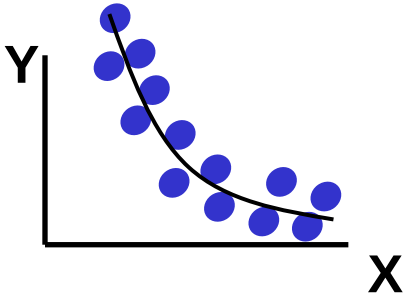




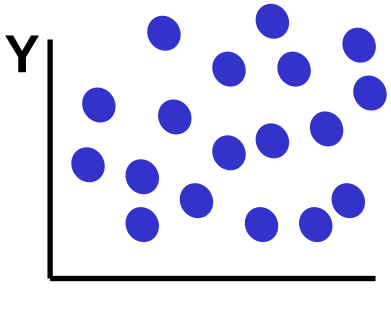
Curvilinear Relationships

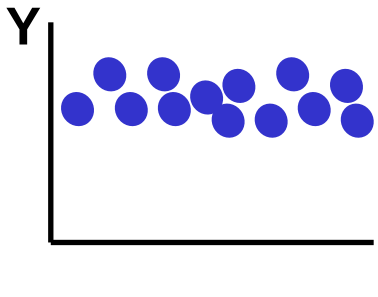






No Relationship





Copyright © Pearson Education Ltd.

Simple Linear Regression Model

DCOVA

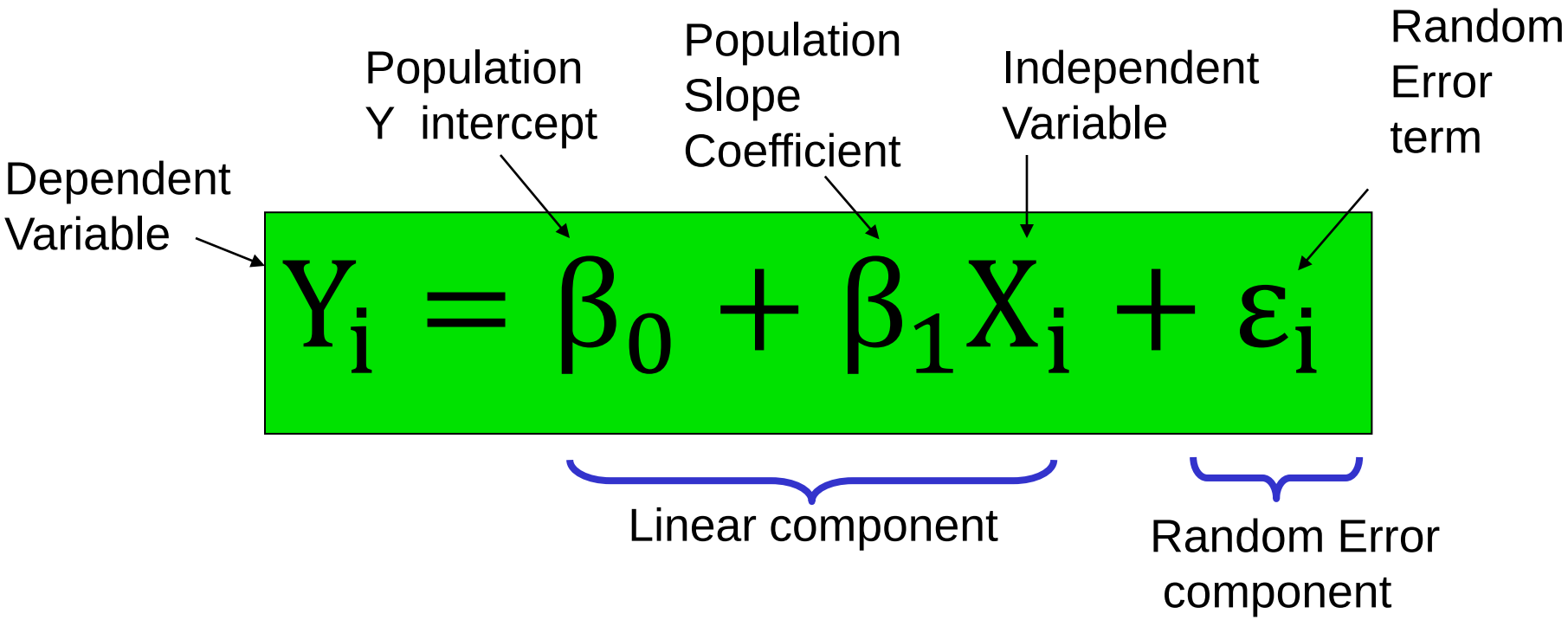
- Only **one** independent variable, X.
- Relationship between X and Y is described by a linear function.
- Changes in Y are assumed to be related to changes in X.

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918 · 2018

Copyright © Pearson Education Ltd.

Simple Linear Regression Model

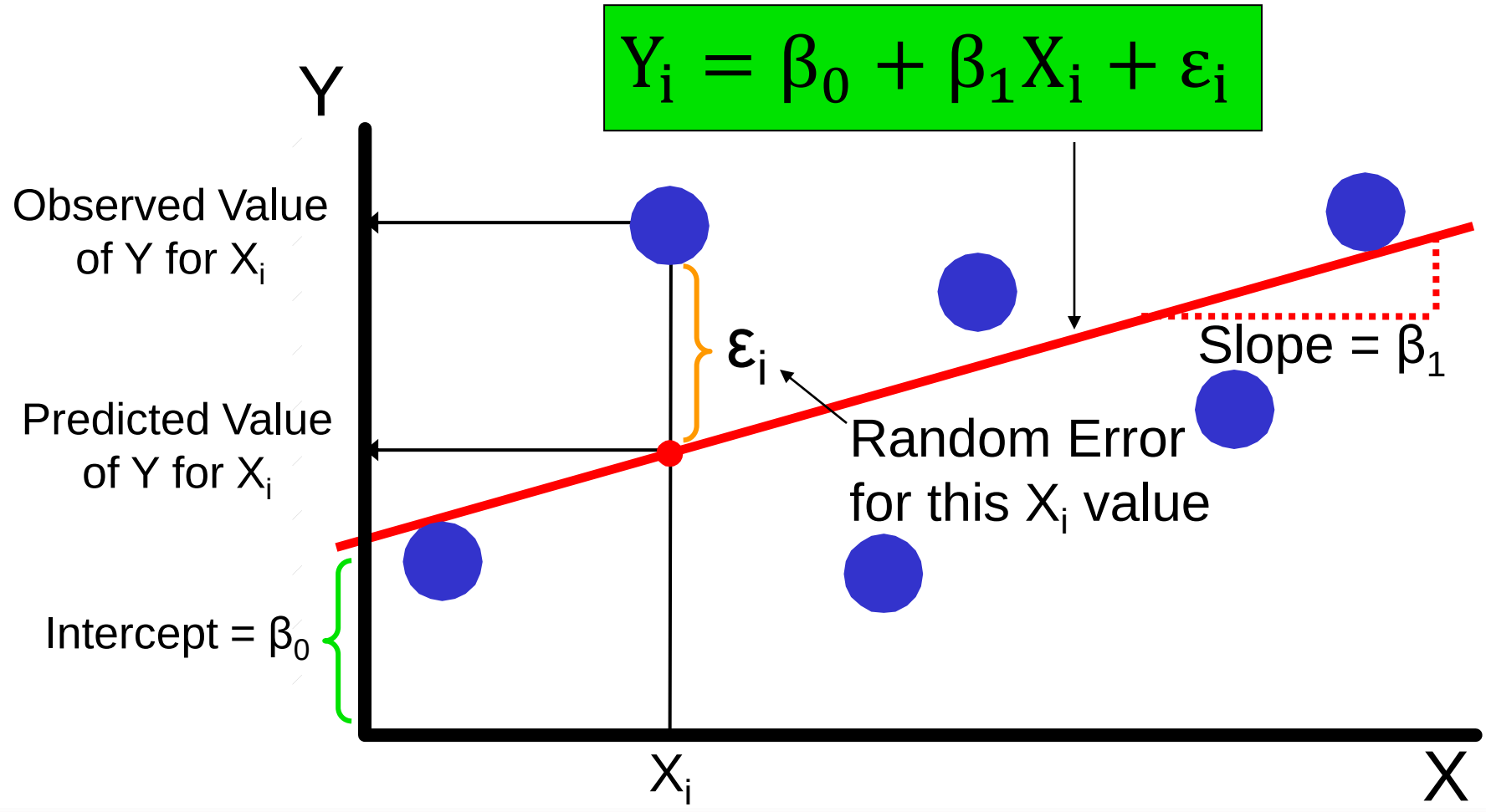
DCOVA



Copyright © Pearson Education Ltd.

Simple Linear Regression Model

DCOVA
(continued)



Copyright © Pearson Education Ltd.

Simple Linear Regression Equation (Prediction Line)

The simple linear regression equation provides an estimate of the population regression line.

Estimated (or predicted) Y value for observation i

Estimate of the regression intercept

Estimate of the regression slope

Value of X for observation i

$$\hat{Y}_i = b_0 + b_1 X_i$$

Copyright © Pearson Education Ltd.

The Least Squares Method

- b_0 and b_1 are obtained by finding the values that minimise the sum of the squared differences between Y and $\hat{Y}$:

$$\min \sum (Y_i - \hat{Y}_i)^2 = \min \sum (Y_i - (b_0 + b_1 X_i))^2$$

- The coefficients b_0 and b_1, and other regression results in this chapter, will be found using Excel.

Copyright © Pearson Education Ltd.

Interpretation of the Slope and the Intercept

- b_0 is the estimated mean value of Y when the value of X is zero.

- b_1 is the estimated change in the mean value of Y as a result of a one-unit increase in X.

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918 · 2018

Copyright © Pearson Education Ltd.

Simple Linear Regression Example

DCOVA

- A real estate agent wishes to examine the relationship between the selling price of a home and its size (measured in square feet).

- A random sample of 10 houses is selected.
  - Dependent variable (Y) = house price in $1,000s.
  - Independent variable (X) = square feet.

Copyright © Pearson Education Ltd.

Simple Linear Regression Example: Data

DCOVA

House Price in $1000s (Y)	Square Feet (X)
245	1,400
312	1,600
279	1,700
308	1,875
199	1,100
219	1,550
405	2,350
324	2,450
319	1,425
255	1,700

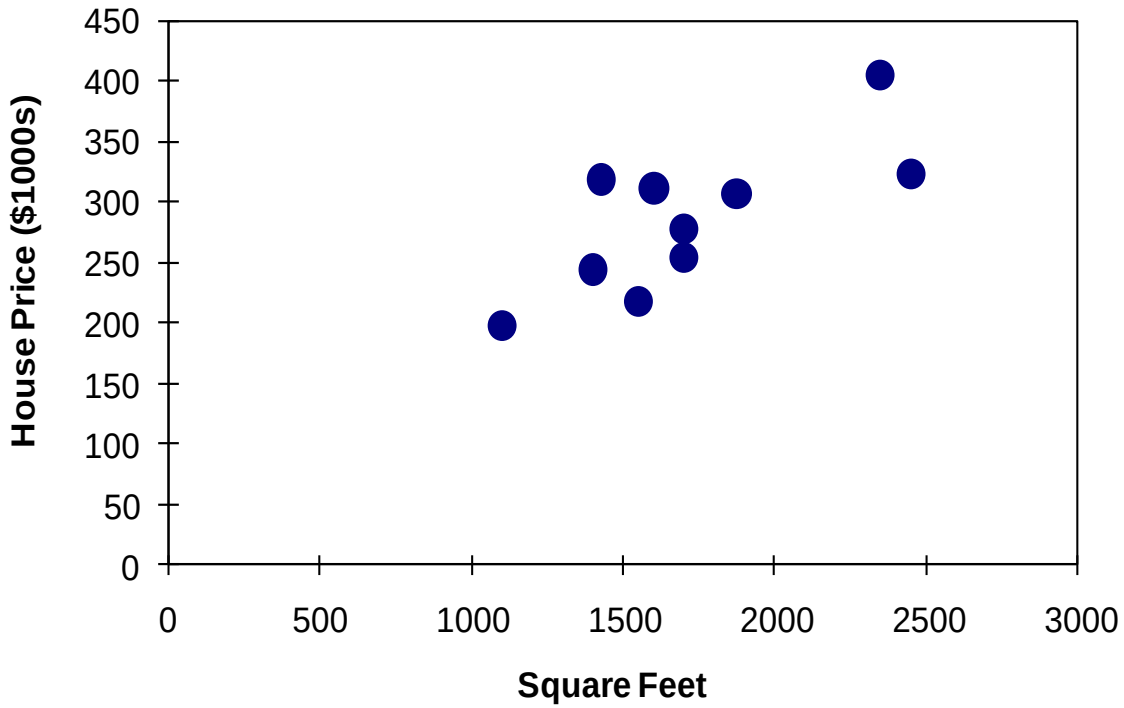
UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918 · 2018

Copyright © Pearson Education Ltd.

Simple Linear Regression Example: Scatter Plot

DCOVA

House price model: Scatter Plot



UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918 · 2018

Copyright © Pearson Education Ltd.

Simple Linear Regression Example: Excel Output

DCOVA

Regression Statistics	
Multiple R	0.76211
R Square	0.58082
Adjusted R Square	0.52842
Standard Error	41.33032
Observations	10

The regression equation is:

$$\widehat{\text{house price}} = 98.24833 + 0.10977\,(\text{square feet})$$

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	1	18934.9348	18934.9348	11.0848	0.01039
Residual	8	13665.5652	1708.1957		
Total	9	32600.5000			

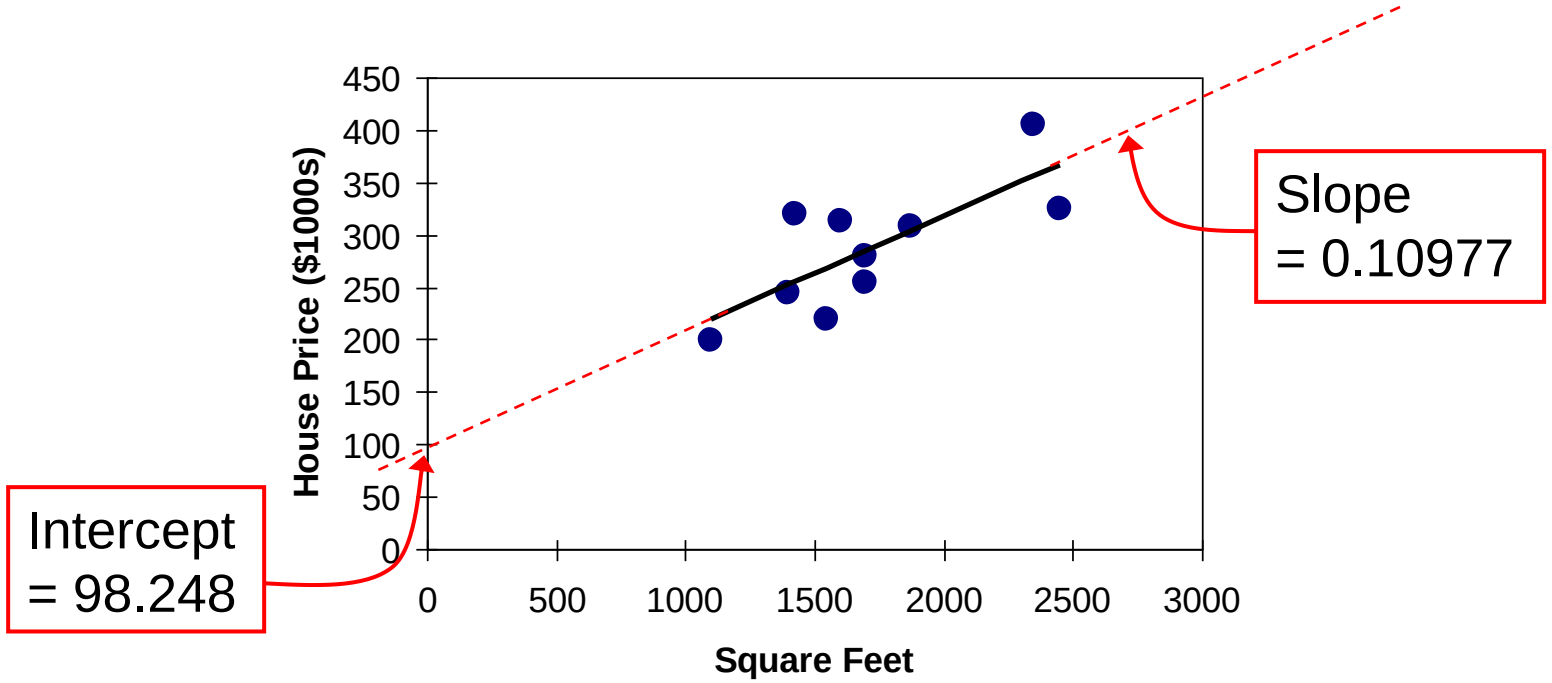
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	98.24833	58.03348	1.69296	0.12892	-35.57720	232.07386
Square Feet	0.10977	0.03297	3.32938	0.01039	0.03374	0.18580

Copyright © Pearson Education Ltd.

Simple Linear Regression Example: Graphical Representation

DCOVA

House price model: Scatter Plot and Prediction Line



$$\widehat{\text{house price}} = 98.24833 + 0.10977\,(\text{square feet})$$

Copyright © Pearson Education Ltd.

Simple Linear Regression Example: Interpretation of b_0

DCOVA

$$\widehat{\text{house price}} = 98.24833 + 0.10977\,(\text{square feet})$$

- b_0 is the estimated mean value of Y when the value of X is zero (if X = 0 is in the range of observed X values)
- Because a house cannot have a square footage of 0, b_0 has no practical application

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918·2018

Copyright © Pearson Education Ltd.

Simple Linear Regression Example: Interpreting b_1

DCOVA

$$\widehat{\text{house price}} = 98.24833 + 0.10977\,(\text{square feet})$$

- b_1 estimates the change in the mean value of Y as a result of a one-unit increase in X.
- Here, $b_1 = 0.10977$ tells us that the mean value of a house increases by 0.10977($1,000) = $109.77, on average, for each additional one square foot of size.

Copyright © Pearson Education Ltd.

Simple Linear Regression Example: Making Predictions

DCOVA

Predict the price for a house with 2,000 square feet:

$$\widehat{\text{house price}} = 98.25 + 0.1098\,(\text{sq.ft.})$$
$$= 98.25 + 0.1098(2{,}000)$$
$$= 317.85$$

The predicted price for a house with 2,000 square feet is 317.85($1,000s) = $317,850

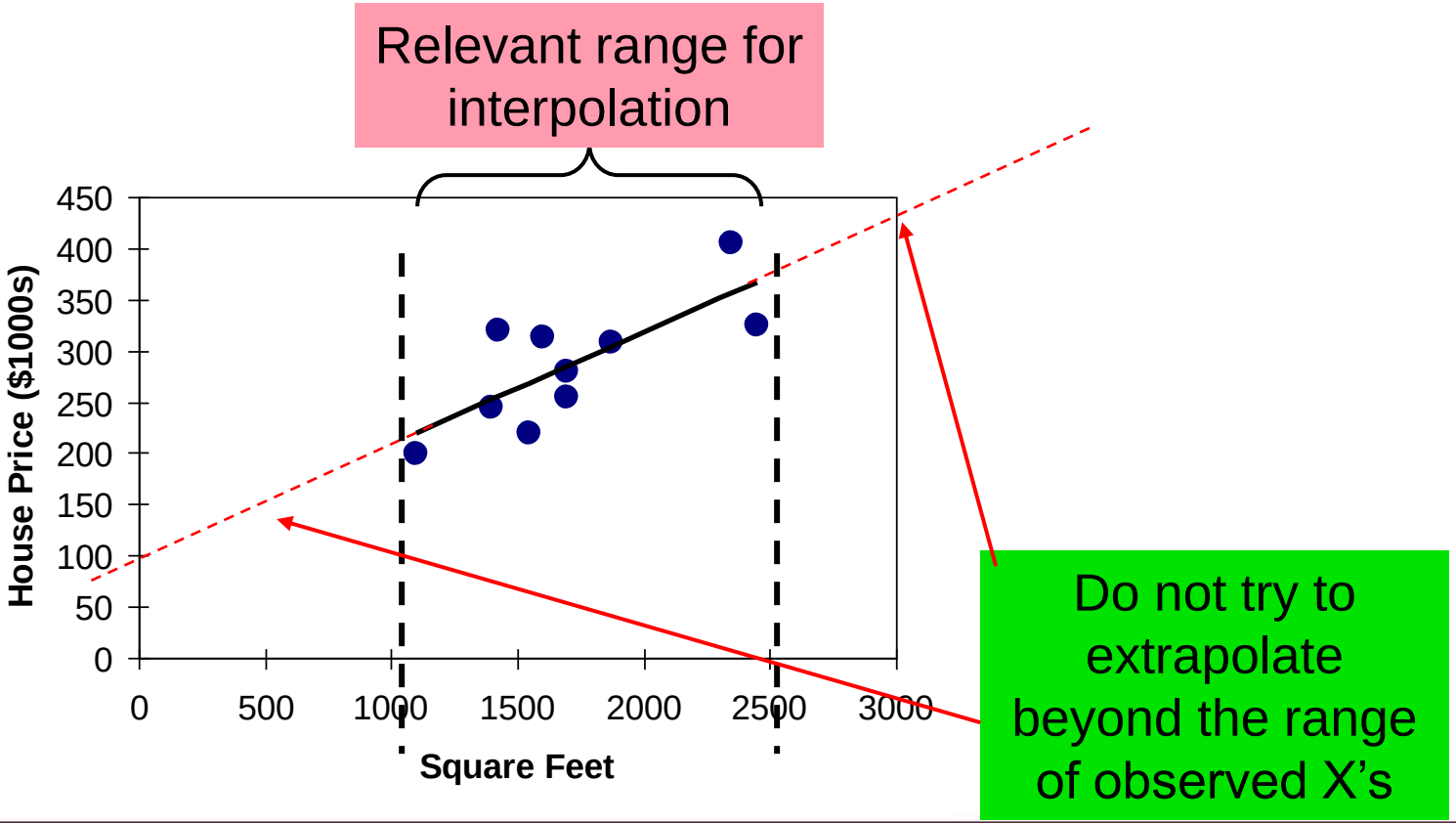
UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918 · 2018

Copyright © Pearson Education Ltd.

Simple Linear Regression Example: Making Predictions

DCOVA

- When using a regression model for prediction, only predict within the relevant range of data



Copyright © Pearson Education Ltd.

Computing b_0 & b_1

For small data sets, b_0 & b_1 can be calculated using a hand calculator

$$b_1 = \frac{SSXY}{SSX} \qquad b_0 = \bar{Y} - b_1\bar{X}$$

Where:

$$SSXY = \sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^{n} X_i Y_i - \frac{(\sum_{i=1}^{n} X_i)(\sum_{i=1}^{n} Y_i)}{n}$$

$$SSX = \sum_{i=1}^{n}(X_i - \bar{X})^2 = \sum_{i=1}^{n} X_i^2 - \frac{(\sum_{i=1}^{n} X_i)^2}{n}$$

$$\bar{Y} = \frac{\sum_{i=1}^{n} Y_i}{n} \quad \text{and} \quad \bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

Copyright © Pearson Education Ltd.

Computing b_0 & b_1 -- Home Prices

DCOVA

House	House Price in $1000s (Y)	Square Feet (X)	X^2	Y^2	XY
1	245	1,400	1,960,000	60,025	343,000
2	312	1,600	2,560,000	97,344	499,200
3	279	1,700	2,890,000	77,841	474,300
4	308	1,875	3,515,625	94,864	577,500
5	199	1,100	1,210,000	39,601	218,900
6	219	1,550	2,402,500	47,961	339,450
7	405	2,350	5,522,500	164,025	951,750
8	324	2,450	6,002,500	104,976	793,800
9	319	1,425	2,030,625	101,761	454,575
10	255	1,700	2,890,000	65,025	433,500
Total	2865	17,150	30,983,750	853,423	5,085,975

$$SSXY = \sum_{i=1}^{n} X_i Y_i - \frac{(\sum_{i=1}^{n} X_i)(\sum_{i=1}^{n} Y_i)}{n} = 5{,}085{,}975 - \frac{(17{,}150)(2{,}865)}{10} = 172{,}500$$

$$SSX = \sum_{i=1}^{n} X_i^2 - \frac{(\sum_{i=1}^{n} X_i)^2}{n} = 30{,}983{,}750 - \frac{17{,}150^2}{10} = 1{,}571{,}500$$

$$\bar{Y} = \frac{\sum_{i=1}^{n} Y_i}{n} = \frac{2{,}865}{10} = 286.5 \quad \text{and} \quad \bar{X} = \frac{\sum_{i=1}^{n} X_i}{n} = \frac{17{,}150}{10} = 1{,}715$$

Copyright © Pearson Education Ltd.

Computing b_0 & b_1 -- Home Prices (Con't)

DCOVA

$$SSXY = \sum_{i=1}^{n} X_i Y_i - \frac{(\sum_{i=1}^{n} X_i)(\sum_{i=1}^{n} Y_i)}{n} = 5{,}085{,}975 - \frac{(17{,}150)(2{,}865)}{10} = 172{,}500$$

$$SSX = \sum_{i=1}^{n} X_i^2 - \frac{(\sum_{i=1}^{n} X_i)^2}{n} = 30{,}983{,}750 - \frac{17{,}150^2}{10} = 1{,}571{,}500$$

$$\bar{Y} = \frac{\sum_{i=1}^{n} Y_i}{n} = \frac{2{,}865}{10} = 286.5 \quad \text{and} \quad \bar{X} = \frac{\sum_{i=1}^{n} X_i}{n} = \frac{17{,}150}{10} = 1{,}715$$

$$b_1 = \frac{SSXY}{SSX} = \frac{172{,}500}{1{,}571{,}500} = 0.1098$$

$$b_0 = \bar{Y} - b_1 \bar{X} = 286.5 - 0.1098 * 1{,}715 = 98.193$$

Copyright © Pearson Education Ltd.

Measures of Variation

DCOVA

- Total variation is made up of two parts:

$$SST = SSR + SSE$$

Total Sum of Squares	Regression Sum of Squares	Error Sum of Squares
$SST = \sum (Y_i - \overline{Y})^2$	$SSR = \sum (\hat{Y}_i - \overline{Y})^2$	$SSE = \sum (Y_i - \hat{Y}_i)^2$

where:

$\overline{Y}$ = Mean value of the dependent variable

Y_i = Observed value of the dependent variable

$\hat{Y}_i$ = Predicted value of Y for the given X_i value

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918·2018

Copyright © Pearson Education Ltd.

Measures of Variation

(continued)

DCOVA

- SST = total sum of squares (Total Variation.)
  - Measures the variation of the Y_i values around their mean $\bar{Y}$.
- SSR = regression sum of squares (Explained Variation.)
  - Variation attributable to the relationship between X and Y.
- SSE = error sum of squares (Unexplained Variation.)
  - Variation in Y attributable to factors other than X.

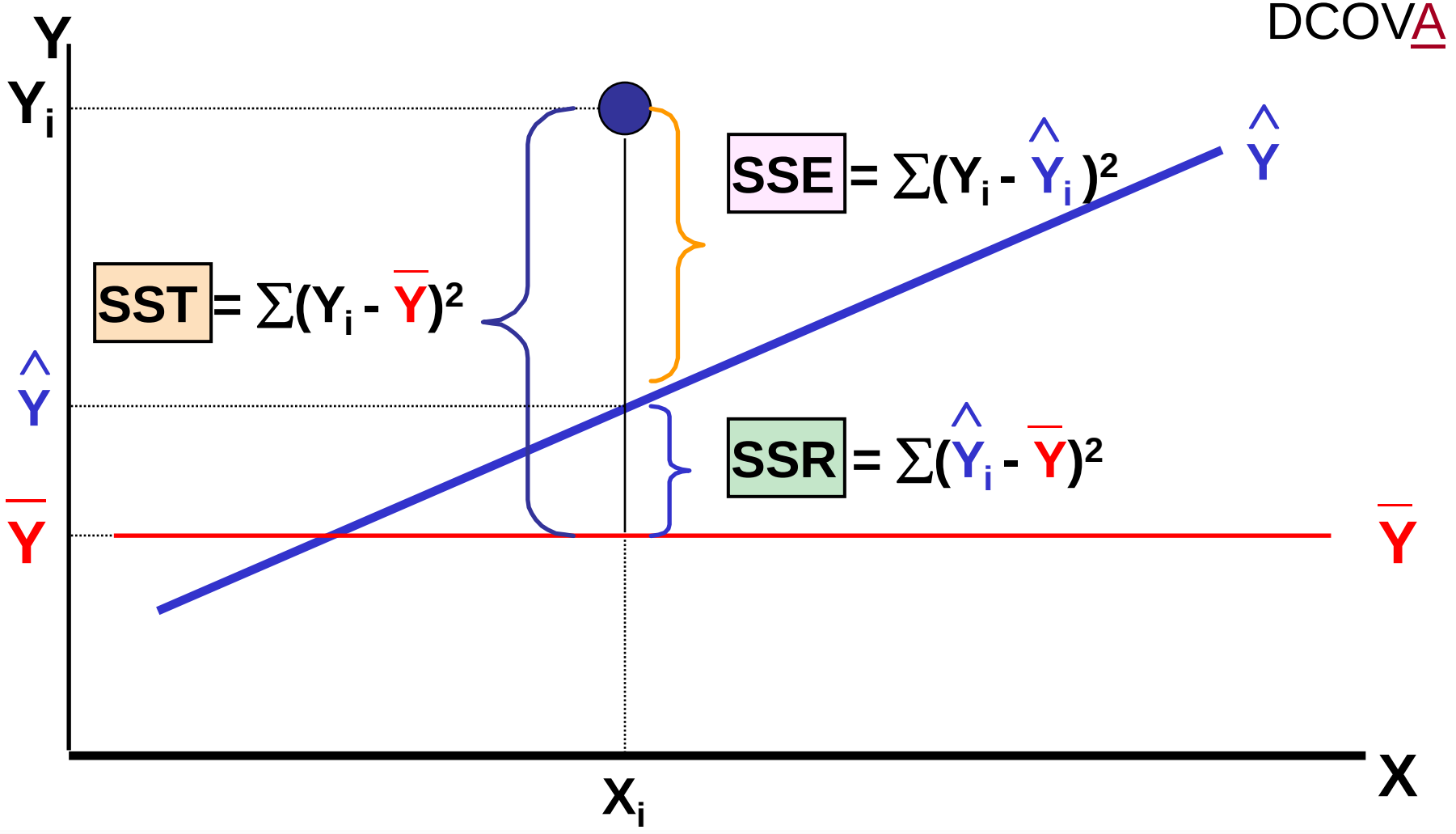
UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918·2018

Copyright © Pearson Education Ltd.

Measures of Variation

(continued)

DCOVA



UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918·2018

Copyright © Pearson Education Ltd.

Excel Output Of The Measures Of Variation

DCOVA

10	ANOVA					
11		*df*	*SS*	*MS*	*F*	*Significance F*
12	Regression	1	18934.9348	18934.9348	11.0848	0.0104
13	Residual	8	13665.5652	1708.1957		
14	Total	9	32600.5000			

SST = SSR + SSE

32,600.5000 = 18,934.9348 + 13,665.5652

Copyright © Pearson Education Ltd.

Coefficient of Determination, r^2

DCOVA

- The coefficient of determination is the portion of the total variation in the dependent variable that is explained by variation in the independent variable.
- The coefficient of determination is also called r-square and is denoted as r^2.

$$r^2 = \frac{SSR}{SST} = \frac{\text{regression sum of squares}}{\text{total sum of squares}}$$

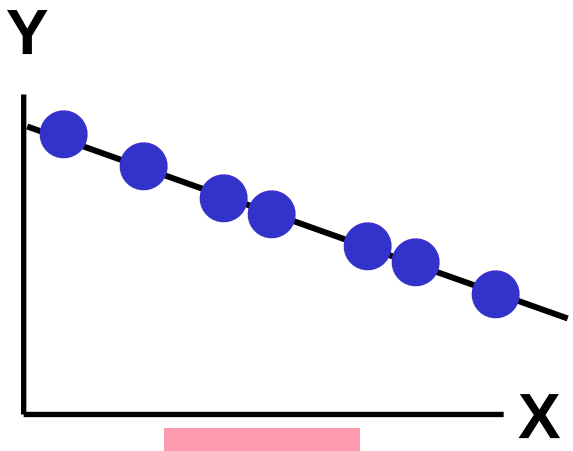
note: $0 \leq r^2 \leq 1$

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918·2018

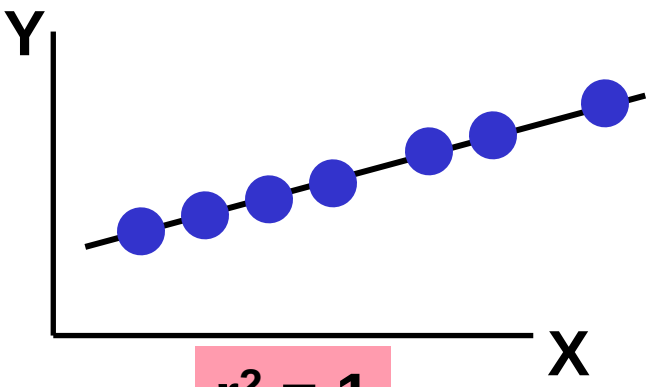
Copyright © Pearson Education Ltd.

Examples of Approximate r^2 Values

DCOVA



$r^2 = 1$



$r^2 = 1$

Perfect linear relationship between X and Y.

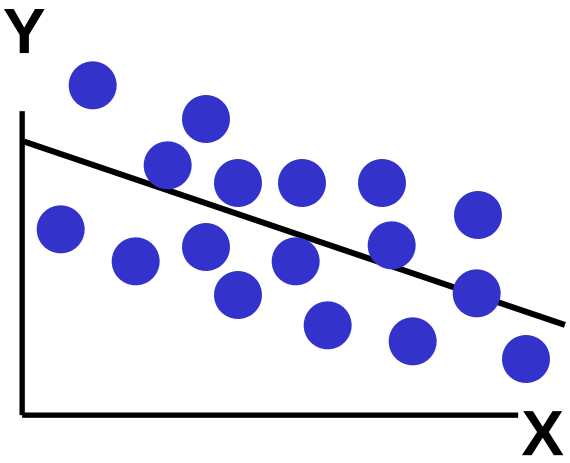
100% of the variation in Y is explained by variation in X.

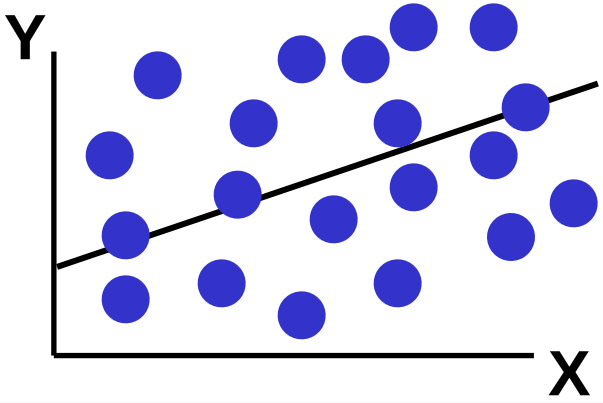
UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918·2018

Copyright © Pearson Education Ltd.

Examples of Approximate r^2 Values

DCOVA





$0 < r^2 < 1$

Weaker linear relationships between X and Y.

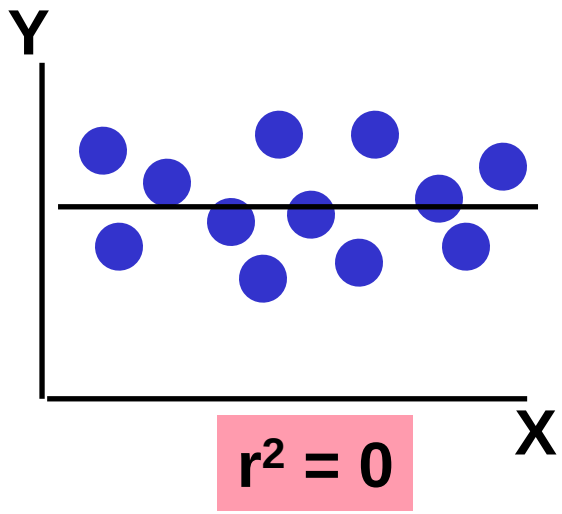
Some but not all of the variation in Y is explained by variation in X.

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918·2018

Copyright © Pearson Education Ltd.

Examples of Approximate r^2 Values

DCOVA



$r^2 = 0$

No linear relationship between X and Y.

The value of Y does not depend on X. (None of the variation in Y is explained by variation in X.)

Copyright © Pearson Education Ltd.

Simple Linear Regression Example: Coefficient of Determination, r^2 in Excel

DCOVA

Regression Statistics	
Multiple R	0.76211
R Square	0.58082
Adjusted R Square	0.52842
Standard Error	41.33032
Observations	10

$$r^2 = \frac{SSR}{SST} = \frac{18{,}934.9348}{32{,}600.5000} = 0.58082$$

58.08% of the variation in house prices is explained by variation in square feet.

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	18934.9348	18934.9348	11.0848	0.01039
Residual	8	13665.5652	1708.1957		
Total	9	32600.5000			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	98.24833	58.03348	1.69296	0.12892	-35.57720	232.07386
Square Feet	0.10977	0.03297	3.32938	0.01039	0.03374	0.18580

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918 · 2018

Copyright © Pearson Education Ltd.

Sum Of Squares Computational Formulae

DCOVA

COMPUTATIONAL FORMULA FOR *SST*

$$SST = \sum_{i=1}^{n}(Y_i - \overline{Y})^2 = \sum_{i=1}^{n}Y_i^2 - \frac{\left(\sum_{i=1}^{n}Y_i\right)^2}{n}$$

COMPUTATIONAL FORMULA FOR *SSR*

$$SSR = \sum_{i=1}^{n}(\hat{Y}_i - \overline{Y})^2$$

$$= b_0\sum_{i=1}^{n}Y_i + b_1\sum_{i=1}^{n}X_iY_i - \frac{\left(\sum_{i=1}^{n}Y_i\right)^2}{n}$$

COMPUTATIONAL FORMULA FOR *SSE*

$$SSE = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 = \sum_{i=1}^{n}Y_i^2 - b_0\sum_{i=1}^{n}Y_i - b_1\sum_{i=1}^{n}X_iY_i$$

Copyright © Pearson Education Ltd.

Standard Error of Estimate

DCOVA

- The standard deviation of the variation of observations around the regression line is estimated by:

$$S_{YX} = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}{n-2}}$$

Where

SSE = error sum of squares

n = sample size

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918·2018

Copyright © Pearson Education Ltd.

Simple Linear Regression Example: Standard Error of Estimate in Excel

DCOVA

Regression Statistics	
Multiple R	0.76211
R Square	0.58082
Adjusted R Square	0.52842
Standard Error	41.33032
Observations	10

$$S_{YX} = \sqrt{\frac{13665.5652}{10-2}} = 41.33032$$

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	1	18934.9348	18934.9348	11.0848	0.01039
Residual	8	13665.5652	1708.1957		
Total	9	32600.5000			

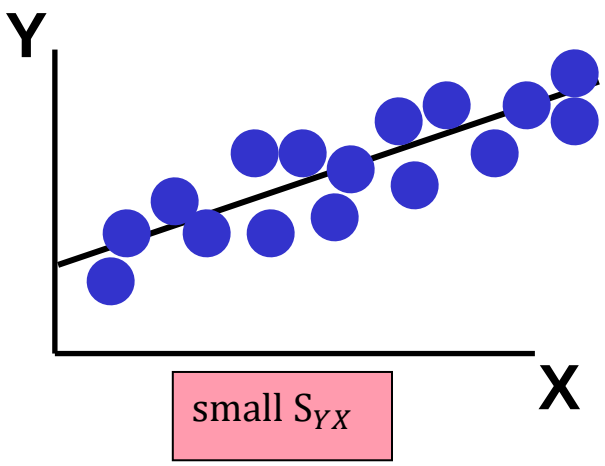
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	98.24833	58.03348	1.69296	0.12892	-35.57720	232.07386
Square Feet	0.10977	0.03297	3.32938	0.01039	0.03374	0.18580

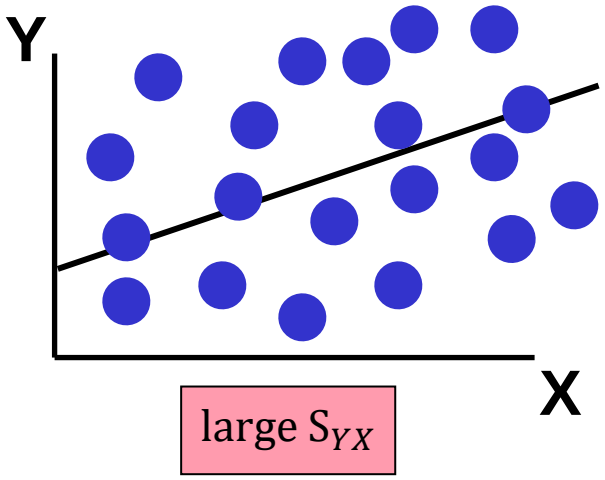
Copyright © Pearson Education Ltd.

Comparing Standard Errors

DCOVA

S_{YX} is a measure of the variation of observed Y values from the regression line.





The magnitude of S_{YX} should always be judged relative to the size of the Y values in the sample data.

i.e., S_{YX} = \$41.33K is moderately small relative to house prices in the \$200K - \$400K range

Copyright © Pearson Education Ltd.

Assumptions of Regression L.I.N.E

DCOVA

- Linearity:
  - The relationship between X and Y is linear.
- Independence of Errors:
  - Error values are statistically independent.
  - Particularly important when data are collected over a period of time.
- Normality of Error:
  - Error values are normally distributed for any given value of X.
- Equal Variance (also called homoscedasticity):
  - The probability distribution of the errors has constant variance.

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918 · 2018

Copyright © Pearson Education Ltd.

Residual Analysis

DCOVA

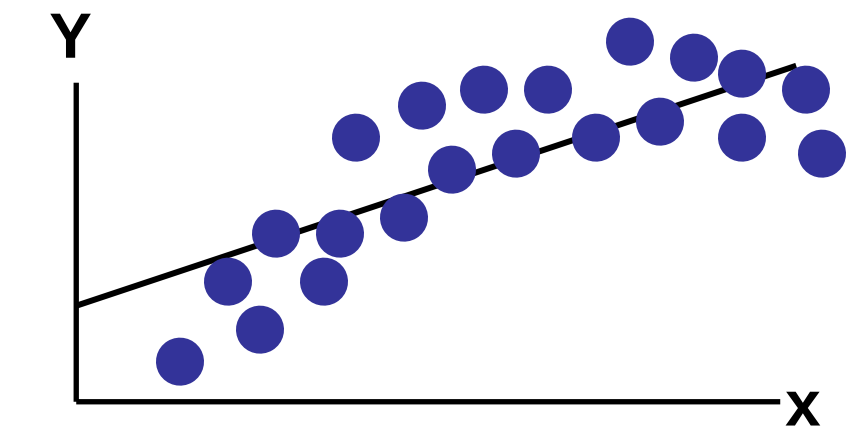
$$e_i = Y_i - \hat{Y}_i$$

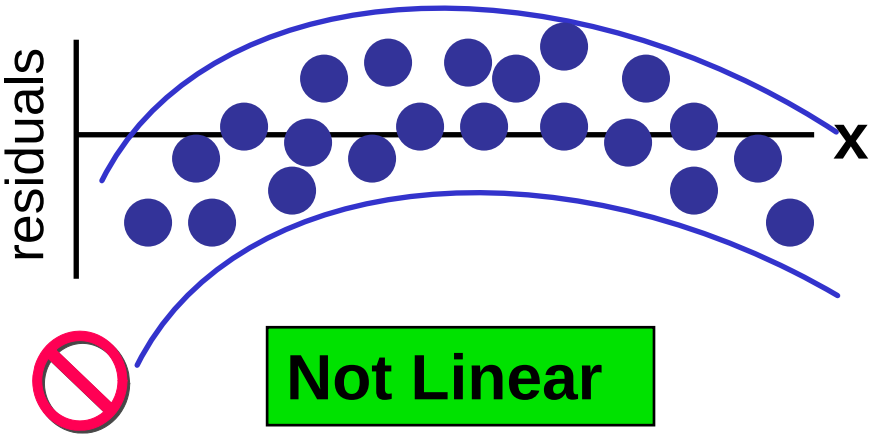
- The residual for observation i, e_i, is the difference between its observed and predicted value.
- Check the assumptions of regression by examining the residuals:
  - Examine for linearity assumption.
  - Evaluate independence assumption.
  - Evaluate normal distribution assumption.
  - Examine for constant variance (homoscedasticity) for all levels of X.
- Graphical Analysis of Residuals
  - Can plot residuals vs. X.

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918·2018
Copyright © Pearson Education Ltd.

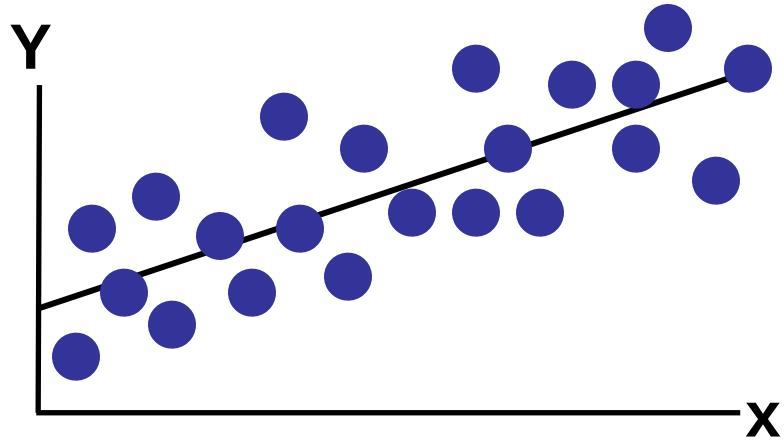
Residual Analysis for Linearity

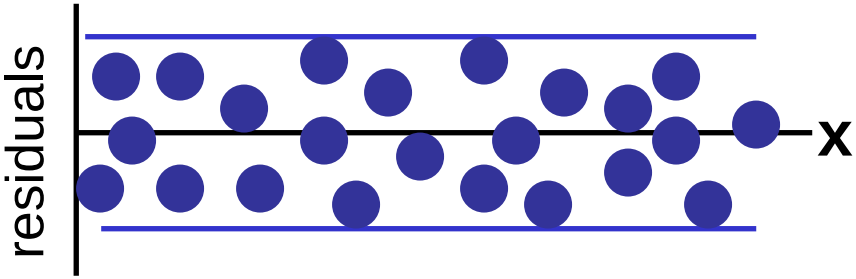
DCOVA





Not Linear





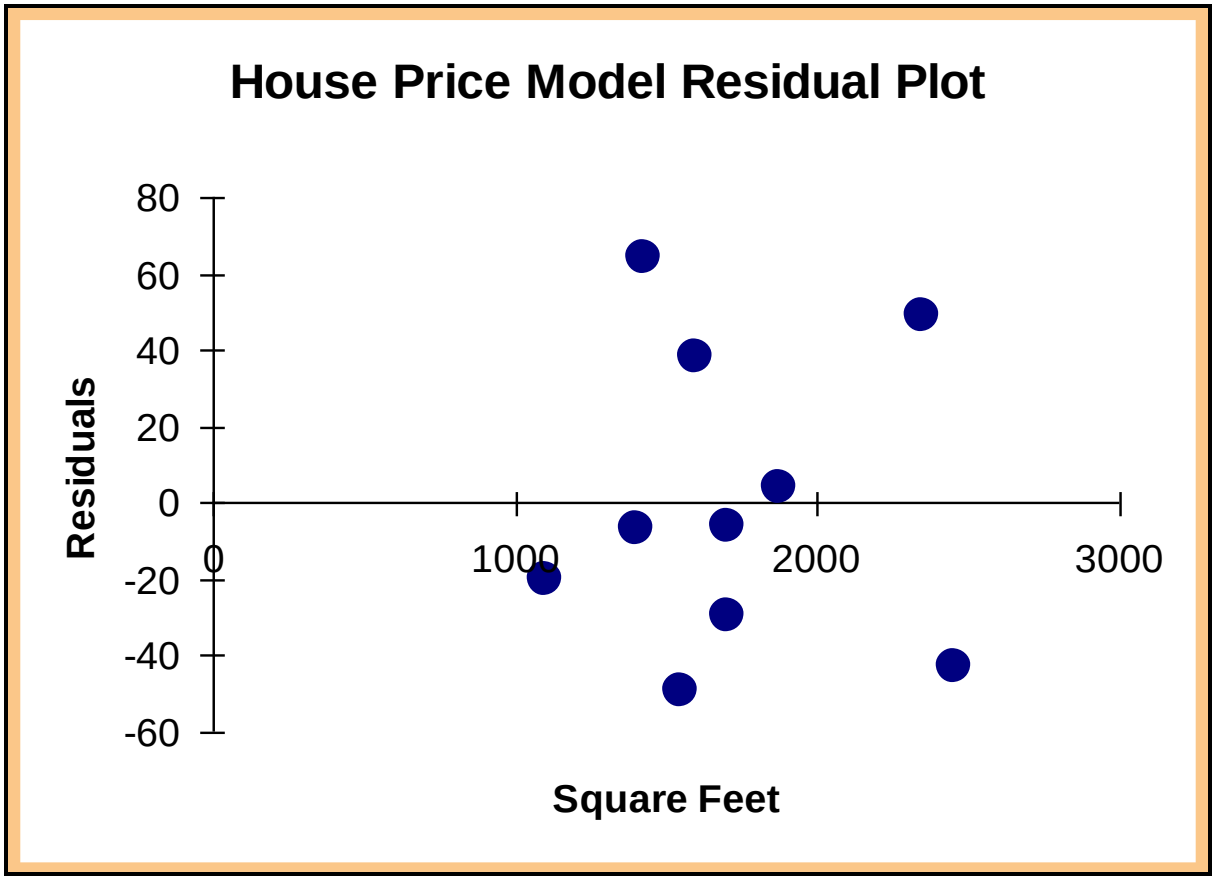
Linear

Copyright © Pearson Education Ltd.

Residual Analysis: Checking For Linearity

DCOVA

RESIDUAL OUTPUT		
	Predicted House Price	*Residuals*
1	251.9232	-6.9232
2	273.8767	38.1233
3	284.8535	-5.8535
4	304.0628	3.9371
5	218.9928	-19.9928
6	268.3883	-49.3883
7	356.2025	48.7975
8	367.1793	-43.1793
9	254.6674	64.3326
10	284.8535	-29.8535



Linear Model Assumption Is Appropriate

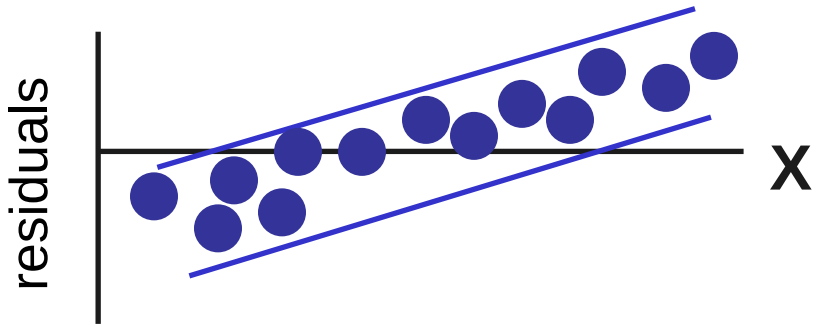
UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918 · 2018
Copyright © Pearson Education Ltd.

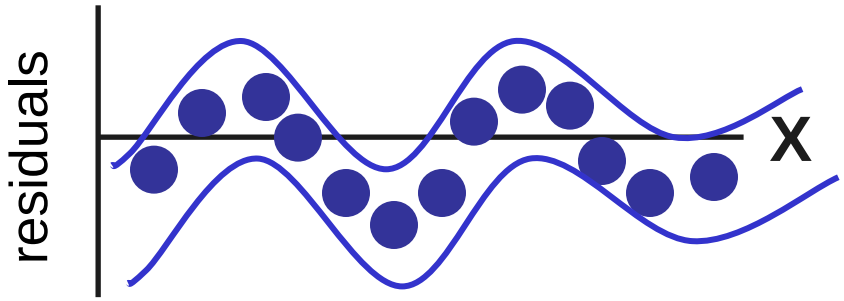
Residual Analysis for Independence

DCOVA

Cyclical Pattern:

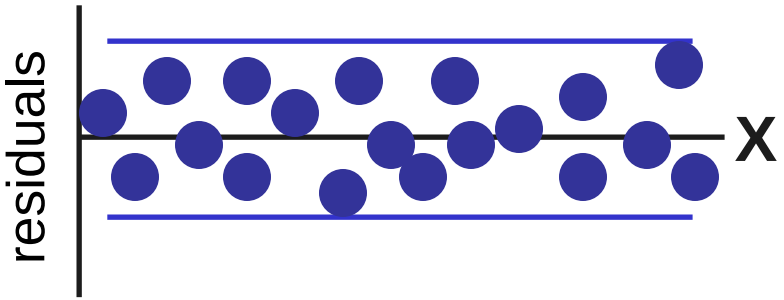
Not Independent





No Cyclical Pattern

Independent



Copyright © Pearson Education Ltd.

Residual Analysis: Checking For Independence (con't.)

DCOVA

Since this data was not collected over time do not need to check for independence.

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918 · 2018

Copyright © Pearson Education Ltd.

Checking for Normality

DCOVA

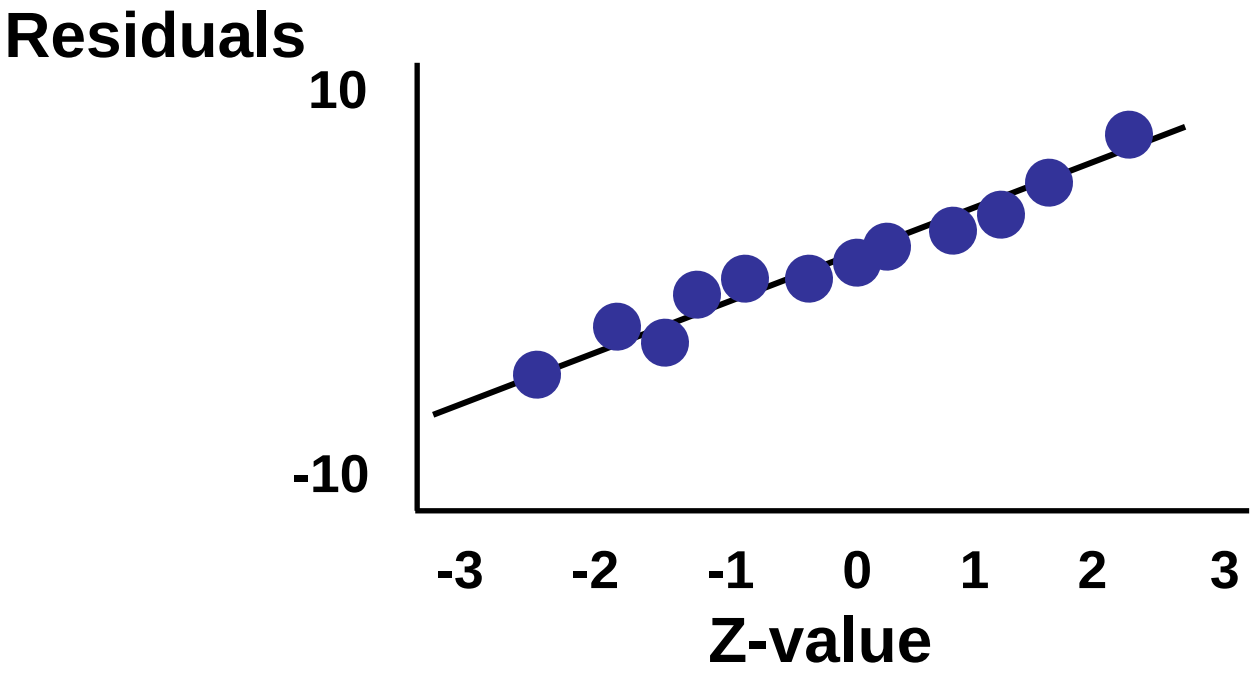
- Examine the Stem-and-Leaf Display of the Residuals.
- Examine the Boxplot of the Residuals.
- Examine the Histogram of the Residuals.
- Construct a Normal Probability Plot of the Residuals.

Copyright © Pearson Education Ltd.

Residual Analysis for Normality

DCOVA

When using a normal probability plot, normal errors will approximately display in a straight line.



UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918 · 2018

Copyright © Pearson Education Ltd.

Residual Analysis for Normality

DCOVA

Normal Probability Plot

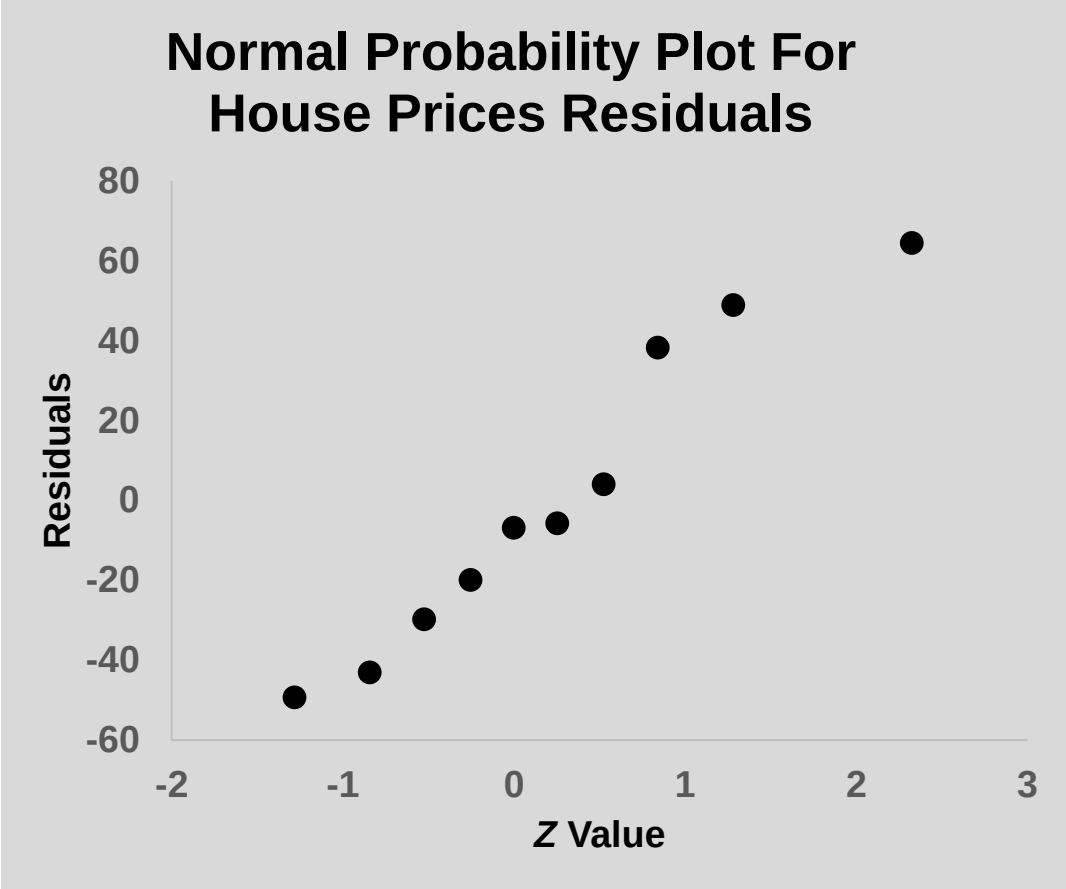
- Sort residuals from small to large: $e_{(1)} < e_{(2)} < \cdots < e_{(n)}$
- Let $p_i = \frac{i}{n+1}, i = 1,2, \ldots, n.$
- Compute corresponding $z_{(i)}, i = 1, \ldots, n$ where $P(Z < z_{(i)}) = p_i.$
- If $e \sim N(0, \sigma^2)$, then $Z = \frac{e-0}{\sigma}$ such that $e = \sigma \cdot Z$ which is a straight line.
- Plot $(z_{(i)}, e_{(i)}), i = 1, \ldots, n$ and visually inspect if the points form a straight line.

Copyright © Pearson Education Ltd.

Residual Analysis: Checking For Normality (con't.)

DCOVA

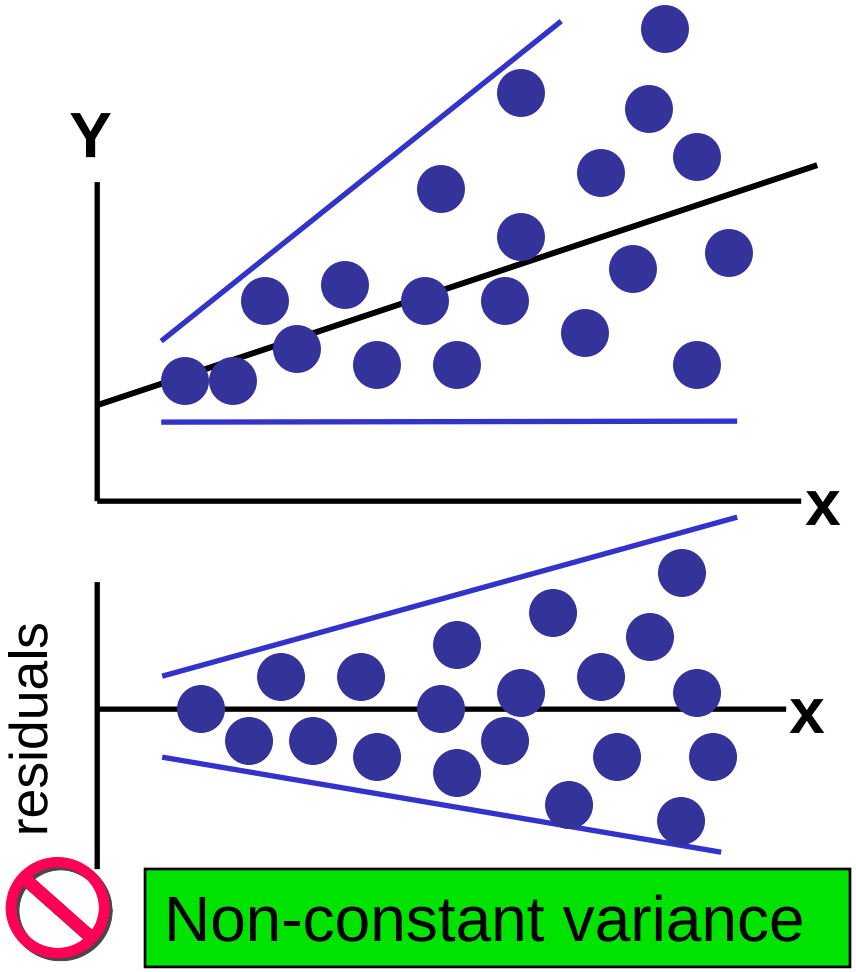
RESIDUAL OUTPUT		
	Predicted House Price	*Residuals*
1	251.9232	-6.9232
2	273.8767	38.1233
3	284.8535	-5.8535
4	304.0628	3.9371
5	218.9928	-19.9928
6	268.3883	-49.3883
7	356.2025	48.7975
8	367.1793	-43.1793
9	254.6674	64.3326
10	284.8535	-29.8535



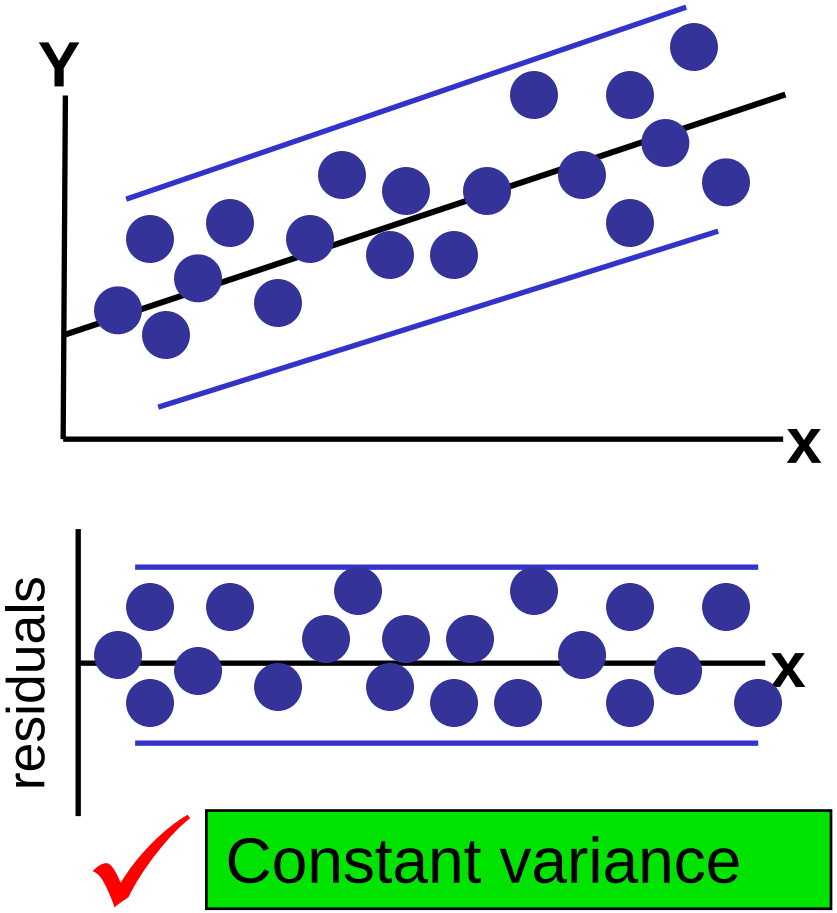
Normality Assumption Is Appropriate

Copyright © Pearson Education Ltd.

Residual Analysis for Equal Variance



Non-constant variance

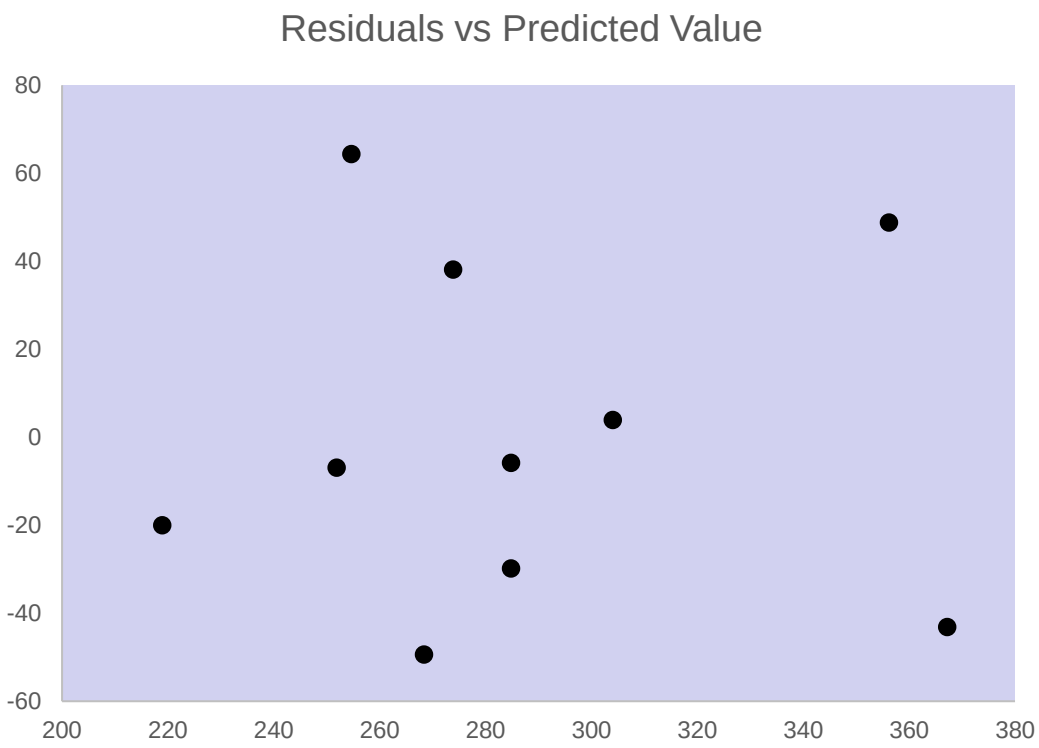


Constant variance

Copyright © Pearson Education Ltd.

Residual Analysis: Checking For Equal Variance (con't.)

RESIDUAL OUTPUT		
	Predicted House Price	*Residuals*
1	251.9232	-6.9232
2	273.8767	38.1233
3	284.8535	-5.8535
4	304.0628	3.9371
5	218.9928	-19.9928
6	268.3883	-49.3883
7	356.2025	48.7975
8	367.1793	-43.1793
9	254.6674	64.3326
10	284.8535	-29.8535



Constant Variance Assumption Is Appropriate

Copyright © Pearson Education Ltd.

Inferences About the Slope

DCOVA

- The standard error of the regression slope coefficient (b_1) is estimated by:

$$S_{b_1} = \frac{S_{YX}}{\sqrt{SSX}} = \frac{S_{YX}}{\sqrt{\sum (X_i - \overline{X})^2}}$$

where:

S_{b_1} = Estimate of the standard error of the slope.

$S_{YX} = \sqrt{\frac{SSE}{n-2}}$ = Standard error of the estimate.

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918·2018

Copyright © Pearson Education Ltd.

Inferences About the Slope: t Test

DCOVA

- t test for a population slope:
  - Is there a linear relationship between X and Y?
- Null and alternative hypotheses:
  - H_0: $\beta_1 = 0$ (no linear relationship)
  - H_1: $\beta_1 \neq 0$ (linear relationship does exist)
- Test statistic :

$$t_{STAT} = \frac{b_1 - \beta_1}{S_{b_1}}$$

where:

b_1 = regression slope coefficient

β_1 = hypothesized slope

S_{b1} = standard error of the slope

$$d.f. = n - 2$$

Copyright © Pearson Education Ltd.

Inferences About the Slope: t Test Example

DCOVA

House Price in $1000s (y)	Square Feet (x)
245	1400
312	1600
279	1700
308	1875
199	1100
219	1550
405	2350
324	2450
319	1425
255	1700

Estimated Regression Equation:

$$\text{house price} = 98.25 + 0.1098\ (\text{sq.ft.})$$

The slope of this model is 0.1098

Is there a relationship between the square footage of the house and its sales price?

Copyright © Pearson Education Ltd.

Inferences About the Slope: t Test Example

DCOVA

$H_0: \beta_1 = 0$

$H_1: \beta_1 \neq 0$

From Excel output:

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	**98.24833**	**58.03348**	**1.69296**	**0.12892**
Square Feet	**0.10977**	**0.03297**	**3.32938**	**0.01039**

b_1

S_{b_1}

$$t_{STAT} = \frac{b_1 - \beta_1}{S_{b_1}} = \frac{0.10977 - 0}{0.03297} = 3.32938$$

Copyright © Pearson Education Ltd.

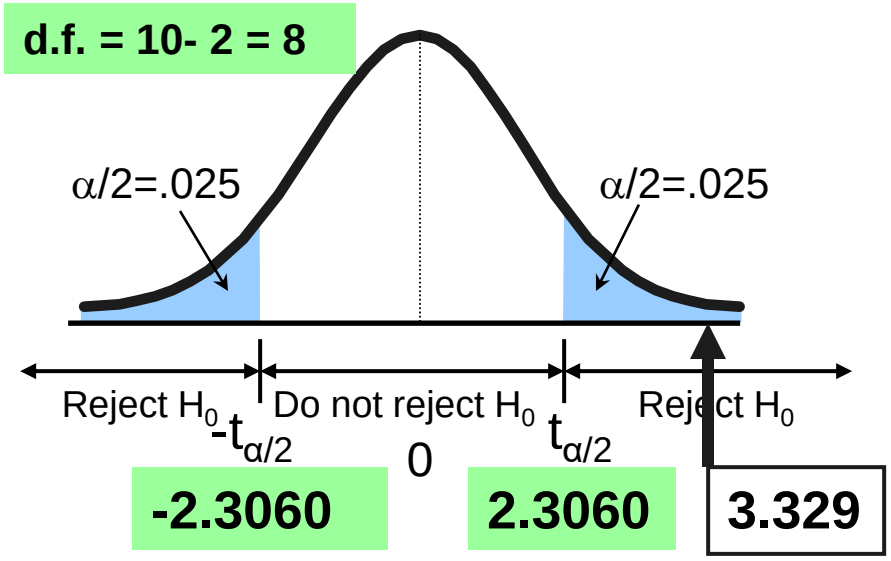
Inferences About the Slope: t Test Example

DCOVA

Test Statistic: $\mathbf{t_{STAT} = 3.329}$

$H_0: \beta_1 = 0$

$H_1: \beta_1 \neq 0$



Decision: Reject H_0.

There is sufficient evidence that square footage affects house price.

Copyright © Pearson Education Ltd.

Inferences About the Slope: t Test Example

DCOVA

$H_0: \beta_1 = 0$

$H_1: \beta_1 \neq 0$

From Excel output:

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	**98.24833**	**58.03348**	**1.69296**	**0.12892**
Square Feet	**0.10977**	**0.03297**	**3.32938**	**0.01039**

p-value

Decision: Reject H_0, since p-value < α.

There is sufficient evidence that square footage affects house price.

Copyright © Pearson Education Ltd.

F Test for The Slope

DCOVA

- F Test statistic: $F_{STAT} = \frac{MSR}{MSE}$

where

$$MSR = \frac{SSR}{1} = SSR$$

$$MSE = \frac{SSE}{n-2}$$

where F_{STAT} follows an F distribution with 1 numerator and (n – 2) denominator degrees of freedom.

Copyright © Pearson Education Ltd.

F-Test for The Slope
Excel Output

DCOVA

Regression Statistics	
Multiple R	0.76211
R Square	0.58082
Adjusted R Square	0.52842
Standard Error	41.33032
Observations	10

$$F_{STAT} = \frac{MSR}{MSE} = \frac{18{,}934.9348}{1{,}708.1957} = 11.0848$$

With 1 and 8 degrees of freedom

p-value for the F-Test

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	1	18934.9348	18934.9348	11.0848	0.01039
Residual	8	13665.5652	1708.1957		
Total	9	32600.5000			

Copyright © Pearson Education Ltd.

F Test for The Slope

(continued)

DCOVA

$H_0: \beta_1 = 0$

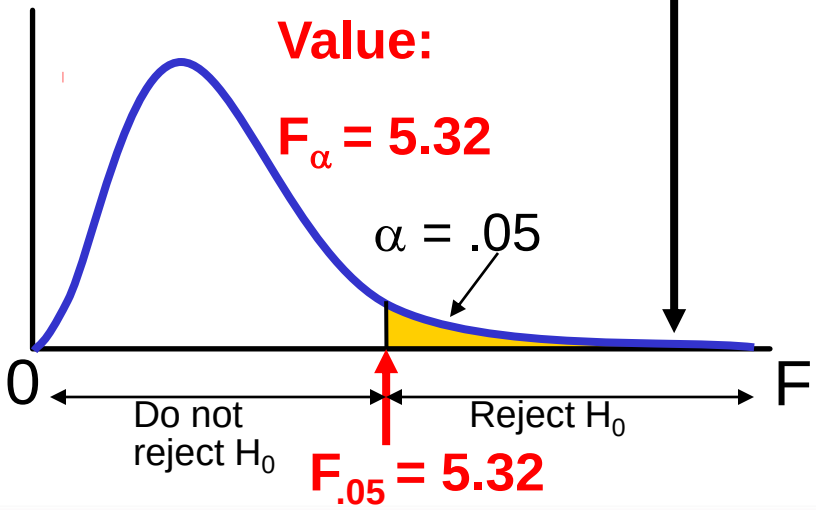
$H_1: \beta_1 \neq 0$

$\alpha = .05$

$df_1 = 1 \qquad df_2 = 8$

Critical Value:

$F_\alpha = 5.32$



Test Statistic:

$$F_{STAT} = \frac{MSR}{MSE} = 11.08$$

Decision:

Reject H_0 at $\alpha = 0.05$.

Conclusion:

There is sufficient evidence that house size affects selling price.

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918·2018

Copyright © Pearson Education Ltd.

Confidence Interval Estimate for the Slope

DCOVA

Confidence Interval Estimate of the Slope:

$$b_1 \pm t_{\alpha/2} S_{b_1}$$

d.f. = n - 2

Excel Printout for House Prices:

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	**98.24833**	**58.03348**	**1.69296**	**0.12892**	**-35.57720**	**232.07386**
Square Feet	**0.10977**	**0.03297**	**3.32938**	**0.01039**	**0.03374**	**0.18580**

At 95% level of confidence, the confidence interval for the slope is (0.0337, 0.1858).

Copyright © Pearson Education Ltd.

Confidence Interval Estimate for the Slope

(continued)

DCOVA

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	**98.24833**	**58.03348**	**1.69296**	**0.12892**	**-35.57720**	**232.07386**
Square Feet	**0.10977**	**0.03297**	**3.32938**	**0.01039**	**0.03374**	**0.18580**

Since the units of the house price variable is $1,000s, we are 95% confident that the average impact on sales price is between $33.74 and $185.80 per square foot of house size

This 95% confidence interval does not include 0.

Conclusion: There is a significant relationship between house price and square feet at the .05 level of significance.

Copyright © Pearson Education Ltd.

t Test for A Correlation Coefficient

DCOVA

- Hypotheses

H_0: $\rho = 0$	(no correlation between X and Y)
H_1: $\rho \neq 0$	(correlation exists)

- Test statistic

$$t_{STAT} = \frac{r - \rho}{\sqrt{\frac{1 - r^2}{n - 2}}}$$

(with n – 2 degrees of freedom)

where

$r = +\sqrt{r^2}$ if $b_1 > 0$

$r = -\sqrt{r^2}$ if $b_1 < 0$

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918 · 2018

Copyright © Pearson Education Ltd.

t-test For A Correlation Coefficient

(continued)

DCOVA

Is there evidence of a linear relationship between square feet and house price at the .05 level of significance?

H_0: $\rho = 0$ (No correlation)

H_1: $\rho \neq 0$ (correlation exists)

$\alpha = .05$, df = 10 - 2 = 8

$$t_{STAT} = \frac{r - \rho}{\sqrt{\frac{1 - r^2}{n - 2}}} = \frac{.762 - 0}{\sqrt{\frac{1 - .762^2}{10 - 2}}} = 3.328$$

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918·2018

Copyright © Pearson Education Ltd.

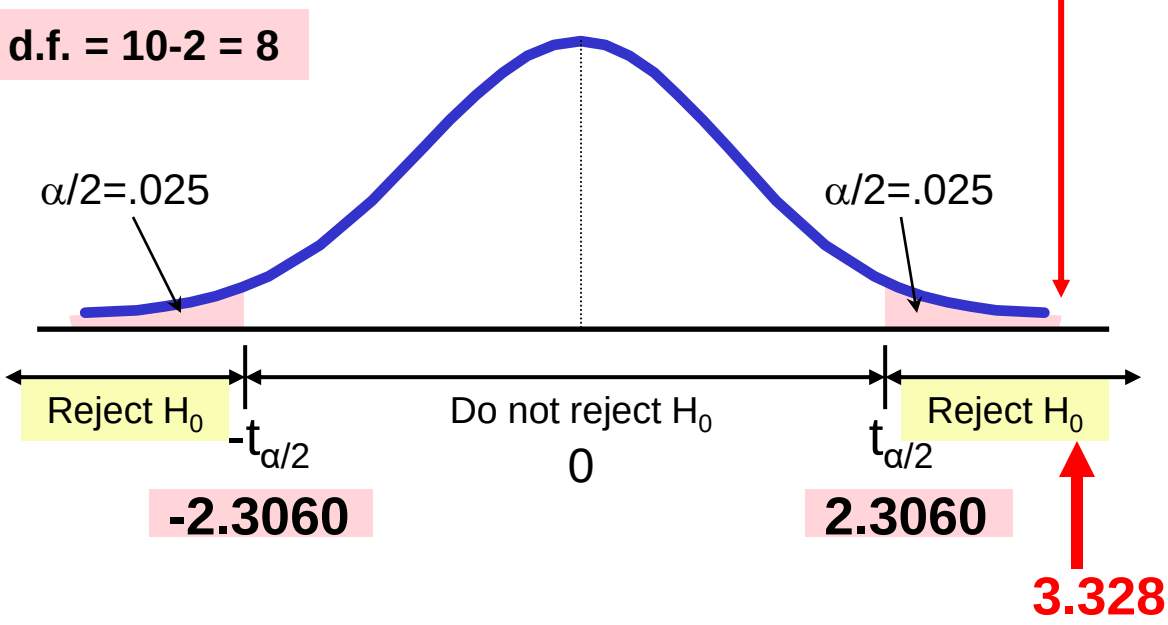
t-test For A Correlation Coefficient

(continued)

DCOVA

$$t_{STAT} = \frac{r - \rho}{\sqrt{\frac{1 - r^2}{n - 2}}} = \frac{.762 - 0}{\sqrt{\frac{1 - .762^2}{10 - 2}}} = 3.328$$

d.f. = 10-2 = 8



Decision:
Reject H_0.

Conclusion:
There **is evidence** of a linear association at the 5% level of significance.

Copyright © Pearson Education Ltd.

Pitfalls of Regression Analysis

DCOVA

- Lacking an awareness of the assumptions of least-squares regression.
- Not knowing how to evaluate the assumptions of least-squares regression.
- Not knowing the alternatives to least-squares regression if a particular assumption is violated.
- Using a regression model without knowledge of the subject matter.
- Extrapolating outside the relevant range.
- Concluding that a significant relationship identified always reflects a cause-and-effect relationship.

Copyright © Pearson Education Ltd.

Seven Steps for Avoiding the Potential Pitfalls

DCOVA

1. Be clear about the problem or goal being investigated and the variables that need to be examined.
2. Construct a scatter plot of X vs. Y to observe the possible relationship.
3. Perform a residual analysis to check the assumptions:
   a. Plot the residuals versus X to check for violations of assumptions such as equal variance.
   b. Use a histogram, stem-and-leaf display, boxplot, or normal probability plot of the residuals to uncover possible non-normality.
   c. Plot the residuals versus time to check for independence. (This step is necessary only if the data are collected over time.)

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918 · 2018

Copyright © Pearson Education Ltd.

Seven Steps for Avoiding the Potential Pitfalls

(continued)

DCOVA

4. If there are violations of the assumptions, use alternative methods to least-squares regression or alternative least-squares models.
5. If there are no violations of the assumptions, carry out tests for the significance of the regression coefficients and develop confidence and prediction intervals.
6. Refrain from making predictions and forecasts outside the relevant range of the independent variable.
7. Remember that the relationships identified in observational studies may or may not be due to cause-and-effect relationships (While causation implies correlation, correlation does not imply causation.)

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918·2018

Copyright © Pearson Education Ltd.

Chapter Summary

In this chapter we discussed:

- The meaning of the regression coefficients b_0 and b_1.

- Understanding the usefulness of regression analysis and how to properly perform regression analysis.

UNIVERSITEIT iYUNIVESITHI STELLENBOSCH UNIVERSITY 100 1918 · 2018

Copyright © Pearson Education Ltd.